

STEP-BY-STEP WALKTHROUGHS

The 10 processes — follow them yourself

Each card tells you **what the process is for**, shows the flow at a glance, then lists numbered steps — **log in as this person, do this** — in order. Switch logins and watch the file travel from start to finish.

1 🌟 Admission — a child joins the school

Open live ↗

9 steps · 5 approvers · the flagship workflow

What it does: takes a family from “we want admission” to “your child has a roll number and a house”. Five school people say YES one after another; anyone's NO sends the file back or ends it.

AO sends link → Parent submits → AO validates docs → Payment → Accounts approves fee → OS: Roll No + House → VP YES → Principal signs → Student created 🎉

1. Log in as `ao@sainik01.edu · Demo@Ao1` → Admissions → send the form link to a parent.
2. Log in as `parent@sainik01.edu · Demo@Parent1` → open the link → fill the form, upload documents, submit.
3. Back to `ao@…` → check documents: good → Validate · bad → send back to parent.
4. Fee in: parent pays online, or `accounts@sainik01.edu · Demo@Accounts1` records it → Fee Approve.
5. Log in as `os@sainik01.edu · Demo@Os1` → Allocate → the child gets a Roll Number + House.
6. Log in as `vp@sainik01.edu · Demo@Vp1` → Approve step 7. (His send-back **erases the roll number!**)
7. Log in as `principal@sainik01.edu · Demo@Principal1` → type your name as signature → CONFIRM. Done — a real student now.

Rule: each step is a locked door — only the right role opens it, and every YES/NO is stored with a remark. The Principal's dashboard shows the whole queue in one screen.

✅ Complete & live: all 9 steps, payment, send-back loops, auto student-master creation.

2 🖍️ Attendance — who came to class today

Open live ↗

daily roll call · instant parent alert

What it does: the teacher marks every child Present / Absent / Late, and the system instantly messages the parents of absent children.

Teacher marks roll → Child marked Absent → AUTO SMS/Email to parent → Below-75% list feeds report cards

1. Log in as `teacher@sainik01.edu · Demo@Teacher1` → Attendance → class 10A → mark the roll → Save.
2. Log in as `parent@sainik01.edu · Demo@Parent1` → Portal → Attendance → the absent mark is already there.
3. Log in as `principal@sainik01.edu · Demo@Principal1` → school-wide registers + the below-75% warning list.

Rule: only teachers, HODs and bosses mark attendance — Warden, Security and Parents can't even see this page. % = (Present+Late+OD) ÷ (Present+Late+OD+Absent); approved leave & Sundays don't count.

✅ Complete & live: bulk marking, no-repeat auto-notifications, registers, analytics.

3 💵 Fees — the school bills, the family pays

Open live ↗

invoice → money → receipt

What it does: creates a fee bill for every student each term, collects money online or in cash, issues a numbered receipt, and lists who hasn't paid.

Accounts: structure → Invoice INV-2026-0001 → Parent pays / cash at office → Receipt RCPT + Paid ✅ → Defaulters list

1. Log in as `accounts@sainik01.edu · Demo@Accounts1` → Fees → set the structure → Generate invoices.
2. Log in as `parent@sainik01.edu · Demo@Parent1` → Portal → Fees → Pay online (or accounts records cash + prints receipt).
3. Back to `accounts@…` → the invoice turns Paid → open the Defaulters list.

Rule: you can never pay more than the balance — the system refuses it. No "fee discount" screen exists; that was decided on purpose.

✅ Complete & live: sequential numbers, balance math, receipt PDFs, online collection.

4 🔐 Gate Pass — a child leaves campus

Open live ↗

the strictest chain: 2 approvals + parent OTP + guard check

What it does: nobody walks out without four checks — the Warden approves, the Principal approves, the parent consents with an OTP, and the guard verifies it at the gate.

Parent requests → Warden YES → Principal YES → OTP sent →
Guard verifies OTP → Checked Out → Returned (+late flag)

1. Log in as `parent@sainik01.edu · Demo@Parent1` → Portal → Gate Passes → request with a time.
2. Log in as `warden@sainik01.edu · Demo@Warden1` → Gate → approve (1st YES).
3. Log in as `principal@sainik01.edu · Demo@Principal1` → approve (2nd YES) → a 6-digit OTP goes to the parent.
4. Log in as `security@sainik01.edu · Demo@Security1` → Gate Desk → enter OTP → Checked Out. On return → Check In (late = flagged).

Rule: 5 wrong OTP tries lock the pass. No OTP = no exit — even with both approvals.

✅ Complete & live: full chain, hashed OTP, QR pass PDF, visitor log, parent notified every turn.

5 🗓️ Staff Leave — a holiday… with a salary effect

Open live ↗

apply → approve → auto-dock over-quota days

What it does: staff apply for leave, a boss approves it, and days beyond the yearly quota are automatically cut from that month's salary.

Staff applies → HOD / HR / Boss decides → Approved / Rejected →

Over-quota days dock payslip

1. Log in as `teacher@sainik01.edu · Demo@Teacher1` → Leaves → apply (Casual / Sick / Earned / Duty / Unpaid).
2. Log in as `hr@sainik01.edu · Demo@Hr1` (or hod) → Approve or Reject — rejection needs a reason.
3. Log in as `principal@sainik01.edu · Demo@Principal1` → Staff → "My payslips" shows the effect on pay.

Rule: nobody approves their own leave — try `hod@…` on his own application and watch it refuse. Quotas: Casual 12 · Sick 8 · Earned 15 · Duty 10 · Unpaid = every day cuts salary.

✅ Complete & live: verified — 3 unpaid days docked ₹3,000 from a ₹30,000 salary.

6 👨 Student Leave — a child stays home

Open live ↗

parent asks → staff decides → attendance pre-filled

What it does: the parent asks for leave for their child; once approved, the attendance sheet pre-fills the child as “Leave” so no teacher marks them absent by mistake.

Parent requests → **Warden / Teacher / HOD decides** →

Attendance & hostel roll-call pre-filled ✨

1. Log in as `parent@sainik01.edu · Demo@Parent1` → Portal → Leaves → request for the child.
2. Log in as `warden@sainik01.edu · Demo@Warden1` (or teacher/hod) → Approve or Reject — parent gets a message either way.
3. Log in as `teacher@sainik01.edu · Demo@Teacher1` → Attendance → the child already shows “Leave”.

Rule: only the person who filed the request can cancel it — and only while it's still Pending.

✅ Complete & live: portal request, overlap check, auto-fill into attendance & roll-call.

7 🖨️ Exams & Report Card — marks in, result out

Open live ↗

the publish gate · plus online CBT tests

What it does: the school announces exams, teachers enter marks, grades and ranks are computed — and only when a boss **PUBLISHES** can the family see the card.

HOD: exam + bands → Teacher marks (Provisional) → HOD/Boss PUBLISHES → Family portal sees it 🎉

1. Log in as `hod@sainik01.edu · Demo@Hod1` → Exams → create the exam + grade bands.
2. Log in as `teacher@sainik01.edu · Demo@Teacher1` → enter marks (card says "Provisional" — family sees nothing yet).
3. Back to `hod@…` (or principal) → Publish the report cards.
4. Log in as `parent@sainik01.edu · Demo@Parent1` → Portal → Results → the card appears.
5. Bonus: log in as `student@sainik01.edu · Demo@Student1` → Tests → attempt the online CBT (server timer, MCQs self-check, teacher grades essays).

Rule: Publish is the gate — before it no parent or student can see any mark. Half-finished results can never leak.

✅ Complete & live: grades, ranks, PDF cards, publish-freeze, CBT with grading queue.

8 💰 Payroll — salary month close

Open live ↗ (Payroll tab)

generate → finalize → paid → staff downloads own slip

What it does: every month HR/accounts create everyone's payslip (tax, EPF, leave deductions already computed), approve the batch, mark it paid — then each staff member sees only their own slip.

HR generates → Draft → Finalize = Generated → Mark Paid → Staff PDF download

1. Log in as `hr@sainik01.edu · Demo@Hr1` (or accounts) → Staff → Payroll → Generate → Finalize → Mark Paid.
2. Log in as `teacher@sainik01.edu · Demo@Teacher1` → Staff → "My payslips" → download the PDF. He sees ONLY his own.

Rule: salary is invisible in the staff directory, and a slip can't be marked Paid before Finalize — no shortcuts.

✅ Complete & live: tax slabs, EPF, professional tax, leave-docking, payslip PDFs.

9 🧭 Timetable & the “teacher is sick” chain

Open live ↗

auto-generated grid · auto-substitutes

What it does: builds the whole school's weekly timetable automatically with zero clashes — and when a teacher is absent, instantly finds a substitute for every free period.

HOD generates → Teacher marked Absent → AUTO substitutes → Substitute emailed + class notice

1. Log in as `hod@sainik01.edu · Demo@Hod1` (or principal) → Timetable → Generate → the full grid appears, no double-bookings.
2. Same login → mark a teacher Absent for today → covers appear for all their periods.
3. Log in as the substitute (e.g. `teacher@sainik01.edu · Demo@Teacher1`) → “My Day” shows the extra class.

Rule: the system checks the substitute's own classes, approved leave and exam duty before assigning — never double-booked. Warden & Security can't see this page at all.

✅ Complete & live: constraint solver, bulk substitution, master/today PDFs, exam duty.

10 🛡️ Discipline — the secret safety files

Open live ↗

anyone can report · only 3 logins can read

What it does: any staff member can report a child-safety or behaviour concern; it becomes a confidential case only three people in the school can open — and every view is logged.

Any staff reports → Open → Case team: Review → Close →

High/Critical = auto-alert to principals

1. Log in as `teacher@sainik01.edu · Demo@Teacher1` → Discipline → report an incident. He can later see only his own report.
2. Log in as `discipline@sainik01.edu · Demo@Discipline1` → open the real case file → review → close.
3. Log in as `principal@sainik01.edu · Demo@Principal1` → re-open a closed case (reason required) or delete.
4. 🧪 Proof test: log in as `schooladmin@sainik01.edu · Demo@Admin1` — otherwise all-powerful — and the case files are NOT there.

Rule: only super_admin, principal and the safeguarding lead read these files. Confidential notes never enter any parent message; every view is written to the audit log.

✅ Complete & live: cases, escalation, row-level confidentiality, full audit trail.

WHO IS WHO

The 15 demo logins

All emails end with **@sainik01.edu**. The live login page has one-click quick-fill buttons — no typing. Parents & students land in the family portal; staff land on the staff dashboard.



SUPER_ADMIN

Software owner

Sees every school. Only one who can create other owners; final word on safety cases.

`admin@sainik01.edu · Admin@123`



BRANCH_ADMIN

Office head

Approves nearly everything inside one school — except the secret safety files.

`schooladmin@sainik01.edu · Demo@Admin1`



PRINCIPAL

Head of school

Final YES on admissions & gate passes; signs with a typed name; closes safety cases.

`principal@sainik01.edu · Demo@Principal1`



VICE_PRINCIPAL

Deputy

Admission step 7 — his send-back erases the roll number. No hostel access.

`vp@sainik01.edu · Demo@Vp1`



OFFICE_SUPERINTENDENT

Senior clerk

Hands out Roll Numbers & Houses; approves the admission form design.

`os@sainik01.edu · Demo@Os1`



ADMISSION_OFFICER

Admission desk

Sends parent links; checks documents; can send a family back to fix them.

`ao@sainik01.edu · Demo@Ao1`



ACCOUNTS

Fee collector

Invoices, receipts, fee approvals, payroll runs, defaulter lists.

`accounts@sainik01.edu · Demo@Accounts1`



HR

Staff manager

Approves staff leave, manages staff records and user logins (never admins).

`hr@sainik01.edu · Demo@Hr1`



HOD

Dept head teacher

Timetable, exam setup, publishes report cards, approves own department's leave.

`hod@sainik01.edu · Demo@Hod1`



TEACHER

Class teacher 10A

Marks attendance, enters marks, decides student leave, runs online tests.

`teacher@sainik01.edu · Demo@Teacher1`



WARDEN

Hostel in-charge

Rooms & roll-call, 1st gate-pass YES, student leave. No academic pages.

`warden@sainik01.edu · Demo@Warden1`



SECURITY

Gate guard

Verifies the OTP at the gate. Smallest access in the system — gate + own payslip.

`security@sainik01.edu · Demo@Security1`



DISCIPLINE_OFFICER

Safeguarding lead

Opens the secret case files; reviews, escalates and closes them.

`discipline@sainik01.edu · Demo@Discipline1`



PARENT

Family

Sees only their linked children: pays fees, requests leave & passes, fills admission forms.

`parent@sainik01.edu · Demo@Parent1`



STUDENT

The child

Sees only their own record; attends classes on screen and writes online tests.

`student@sainik01.edu · Demo@Student1`

ONE GLANCE

Who approves what — and what a NO does

THING	TRAVEL PATH	FINAL YES	NO → WHAT HAPPENS	WHO ACTS (DEMO LOGIN)
Admission template	AO drafts → sends	**OS** publishes	**OS rejects** → back to Draft	ao@… → os@…
Admission application	AO → Accounts → OS → VP → Principal	**Principal signs**	**VP send-back** erases roll+house · Reject = dead	ao@… accounts@… os@… vp@… principal@…
Fee payment	Parent/office pays	**Accounts receipt**	**over-balance refused**	parent@… or accounts@…
Gate pass	Warden → Principal → gate	**both + OTP**	**reject at either stage**; 5 wrong OTPs lock	parent@… warden@… principal@… security@…
Staff leave	Staff applies	**HOD / HR / Boss**	**reject + reason**; never self	teacher@… applies; hod@…/hr@… approves
Student leave	Parent / staff files	**Warden / Teacher / HOD / Boss**	**reject**; filer can Cancel	parent@… files; warden@…/teacher@… decides
Report card	Teacher marks	**HOD/Boss publishes**	**unpublished = family can't see**	teacher@… marks; hod@… publishes; parent@… views
CBT test	Teacher drafts	**publishes**	**no delete after attempts**	teacher@… sets; student@… attempts
Timetable / cover	Teacher requests	**HOD/Boss assigns**	**conflict = refused with reasons**	teacher@… requests; hod@… assigns
Payroll	HR/Accounts generate	**finalize → mark paid**	**Draft can't be Paid**	hr@… or accounts@…; teacher@… views own slip
Discipline case	Any staff reports	**case team closes;**	**delete: super_admin/principal**	any staff; discipline@…; principal@…

THING	TRAVEL PATH	FINAL YES	NO → WHAT HAPPENS	WHO ACTS (DEMO LOGIN)
		Principal reopens		
User account	Admin/HR creates	forced password change at birth	only super_admin makes super_admin; no self-disable	admin@… / hr@… / schooladmin@…

THE FINE PRINT, MADE SIMPLE

The 4 rules that explain the whole system



Right person, right door

Screens and buttons appear only for roles that need them — and the server checks again on every action, so nothing can be faked.



Every NO leaves a note

Approvals and rejections store who did it, when, and why. The audit log can replay any file's whole life.



Families live in a bubble

A parent sees only their children; a student only themselves. One school can never see another school's data.



The system tells the family

Absence, gate pass, fee receipt, admission step, leave result — a message reaches the parent automatically at every turn.

🌐 Live app: genesislumia-erp-demo.vercel.app · Login with one-click demo roles

Deep-dive story with diagrams: **docs/FLOW_AND_ROLES_GUIDE.md** · Source of truth: backend/src/auth/roles.ts, admissions/workflow.ts, prisma/seed.ts

Verified 2026-10-25 — all 17 linked pages returned HTTP 200.